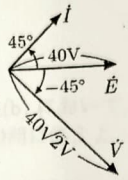
17.6　例題 17.4 と同じ回路で，ω が半分である．例題 17.4 の結果と比較してみよう．

$$\dot{I} = \frac{\dot{E}}{R + j\omega L + \frac{1}{j\omega C}} = \frac{40}{10 + j(10 - 20)}$$

$$= \frac{4}{1 - j} = \frac{4}{\sqrt{2}\angle -45^\circ} = 2\sqrt{2}\angle 45^\circ \text{ [A]}$$

$$\dot{V} = \frac{\dot{I}}{j\omega C} = \frac{2\sqrt{2}\angle 45^\circ}{0.05\angle 90^\circ} = 40\sqrt{2}\angle -45^\circ \text{ [V]}$$



17.7

(1) S_1, S_2 開放時，$\dot{I}$ と $\dot{E}$ が同相ゆえ，インピーダンス $\dot{Z}$ = 実数 R，リアクタンス $\omega L - \dfrac{1}{\omega C} = 0$.

$$\therefore R = Z = \frac{E}{I} = \frac{150}{5} = 30 \text{ [Ω]}$$

S_2 短絡時，$Z = \sqrt{R^2 + (\omega L)^2} = \dfrac{E}{I} = \dfrac{150}{3} = 50.$

$R = 30$ ゆえ $\omega L = 40 \rightarrow L = 0.4$ [H] $= 400$ [mH]

$$\omega L = \frac{1}{\omega C} \rightarrow C = \frac{1}{\omega^2 L} = \frac{1}{4000} \text{ [F]} = 250 \text{ [}\mu\text{F]}$$

(2) $$I = \frac{E}{\sqrt{R^2 + (\frac{1}{\omega C})^2}} = \frac{E}{\sqrt{R^2 + (\omega L)^2}} = 3 \text{ [A]}$$

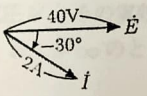
(3) $$I = \frac{E}{R} = 5 \text{ [A]}$$

17.8　R, L, C のインピーダンスはそれぞれ，$10\sqrt{3}$, $j15$, $-j5$ [Ω] となるから，$\dot{Z} = 10\sqrt{3} + j15 - j5$

$= 10\sqrt{3} + j10 = 20\angle 30^\circ$ [Ω].

$$\dot{I} = \frac{40}{20\angle 30^\circ} = 2\angle -30^\circ \text{ [A]}$$

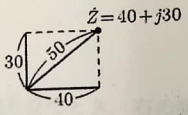
$I = |\dot{I}| = 2$ [A]　$\theta = -30^\circ$



17.9　$\dot{Z} = 40 + j30$ [Ω] だから，

$Z = |\dot{Z}| = \sqrt{40^2 + 30^2} = 50$ [Ω].

$$I = \frac{E}{Z} = \frac{100}{50} = 2 \text{ [A]}$$



17.10　インピーダンス $\dot{Z} = 30 + j10$ [Ω] だから，

アドミタンス $\dot{Y} = \dfrac{1}{30 + j10} = \dfrac{30 - j10}{(30 + j10)(30 - j10)}$

$$= \frac{30 - j10}{30^2 + 10^2} = 0.03 - j0.01 \text{ [S]} = 30 - j10 \text{ [mS]}$$

$\therefore G = 30$ [mS]　$B = -10$ [mS]